

Math 121B — Linear Algebra II

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Winter 2026

Review of 121A

We begin by briefly recalling a few basic notions and notations.

Vector Spaces Bold-face $\mathbf{v}$ denotes a *vector* in a *vector space* V over a *field* $\mathbb{F}$. A vector space is closed under *vector addition* and *scalar multiplication*

$$\forall \mathbf{v}_1, \mathbf{v}_2 \in V, \forall \lambda_1, \lambda_2 \in \mathbb{F}, \quad \lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 \in V \quad (*)$$

Example. Make sure you believe $(*)$ and everything that follows for the vector space

$$V = \mathbb{R}^2 = \{x\mathbf{i} + y\mathbf{j} : x, y \in \mathbb{R}\} = \left\{\begin{pmatrix} x \\ y \end{pmatrix} : x, y \in \mathbb{R}\right\}$$

Since the field $\mathbb{F} = \mathbb{R}$ is the real numbers, this is often referred to as a *real* vector space.

Linear Combinations and Spans Let $\beta \subseteq V$ be a subset of a vector space V over $\mathbb{F}$. A *linear combination* of vectors in β is a *finite* sum

$$\lambda_1 \mathbf{v}_1 + \cdots + \lambda_n \mathbf{v}_n$$

where $\lambda_j \in \mathbb{F}$ and $\mathbf{v}_j \in \beta$. The *span* of β (over $\mathbb{F}$) is the *subspace* of V comprising all linear combinations:

$$\text{Span}_{\mathbb{F}} \beta = \{\lambda_1 \mathbf{v}_1 + \cdots + \lambda_n \mathbf{v}_n : \lambda_j \in \mathbb{F}, \mathbf{v}_j \in \beta\} \leq V$$

Bases and Co-ordinates A set $\beta \subseteq V$ is a *basis* of V if it has two properties:

1. *Linear Independence.* Any linear combination yielding the zero vector is trivial: for distinct $\mathbf{v}_j \in \beta$,

$$\lambda_1 \mathbf{v}_1 + \cdots + \lambda_n \mathbf{v}_n = \mathbf{0} \implies \forall j, \lambda_j = 0$$

2. *Spanning Set.* $V = \text{Span}_{\mathbb{F}} \beta$: every vector in V is a (*finite!*) linear combination of elements of β ,

$$\forall \mathbf{v} \in V, \exists \lambda_j \in \mathbb{F} \text{ for which } \mathbf{v} = \lambda_1 \mathbf{v}_1 + \cdots + \lambda_n \mathbf{v}_n$$

Theorem. β is a basis of $V \iff$ every $\mathbf{v} \in V$ is a unique linear combination of elements of β . The coefficients of $\mathbf{v}$ in this linear combination are its **co-ordinates** with respect to β .

The cardinalities of all basis sets are equal; this value is the **dimension** $\dim_{\mathbb{F}} V$ of the vector space.

It is common to define vector spaces in terms of a basis. For instance, $\mathbb{R}^2$ is defined as the real vector space with basis $\beta = \{\mathbf{i}, \mathbf{j}\} = \left\{\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}\right\}$: this β is known as the *standard basis* of $\mathbb{R}^2$.

Examples. Throughout, let $\mathbb{F}$ be an arbitrary field.¹

1. $\mathbb{F}^n = \text{Span}_{\mathbb{F}}\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ is a vector space of dimension n , defined in terms of its standard basis: $\mathbf{e}_k$ is the zero-vector except for a single 1 in the k^{th} entry.

$$\mathbf{e}_k = \begin{pmatrix} \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \end{pmatrix}$$

The notation $\mathbf{i}, \mathbf{j}, \mathbf{k}$ is a subscript-avoiding shorthand for $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ whenever $n \leq 3$.

2. $P_n(\mathbb{F}) = \text{Span}_{\mathbb{F}}\{1, x, x^2, \dots, x^n\}$ is the space of degree $\leq n$ polynomials with real coefficients.
Warning: $\dim_{\mathbb{F}} P_n(\mathbb{F}) = n + 1$.

Suppose $n = 2$ and $\mathbb{F} = \mathbb{R}$. Every polynomial is a unique linear combination $p(x) = a + bx + cx^2$ with respect to the standard basis $\beta = \{1, x, x^2\}$.

The real numbers a, b, c are the *co-ordinates* of p . The *co-ordinate vector* $[p]_{\beta}$ of p with respect to β is the column vector of these co-ordinates.

$$[p]_{\beta} = \begin{pmatrix} a \\ b \\ c \end{pmatrix} \in \mathbb{R}^3$$

3. $P(\mathbb{F}) = \text{Span}_{\mathbb{F}}\{1, x, x^2, \dots\}$ is the space of polynomials with coefficients in $\mathbb{F}$. This has countably infinite dimension. Note that a polynomial is a *finite sum*: power series are not polynomials!
4. $C(\mathbb{R})$ is the real space of continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$. Its dimension is uncountably infinite. Continuity also makes sense over a field in which limits make sense: e.g. $C(\mathbb{C})$.

Linearity and Linear Maps A function $T : V \rightarrow W$ between vector spaces V, W over the *same field* $\mathbb{F}$ is ($\mathbb{F}$ -)linear if it respects the linearity properties of both spaces

$$\forall \mathbf{v}_1, \mathbf{v}_2 \in V, \forall \lambda_1, \lambda_2 \in \mathbb{F}, \quad T(\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2) = \lambda_1 T(\mathbf{v}_1) + \lambda_2 T(\mathbf{v}_2)$$

We write $\mathcal{L}(V, W)$ for the set (vector space!) of linear maps from V to W , shortened to $\mathcal{L}(V)$ when $V = W$. An *isomorphism* is an invertible/bijective linear map.

Theorem. If $\dim_{\mathbb{F}} V = n$ and β is a basis of V , then the co-ordinate map $\mathbf{v} \mapsto [\mathbf{v}]_{\beta}$ is an isomorphism of vector spaces $V \rightarrow \mathbb{F}^n$.

Matrices and Linear Maps Matrix multiplication is a linear map together with a choice of bases.

Example. The matrix $A = \begin{pmatrix} 2 & -1 & -4 \\ 0 & -1 & 3 \end{pmatrix}$ defines a linear map $L_A : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ (left-multiplication by A):

$$L_A \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 & -1 & -4 \\ 0 & -1 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2x - y - 4z \\ -y + 3z \end{pmatrix}$$

Conversely, the linear map *defines* the matrix A : we recover the columns of A by feeding the standard basis vectors $\mathbf{i}, \mathbf{j}, \mathbf{k}$ (of $\mathbb{R}^3$) to the linear map.

$$\begin{pmatrix} 2 \\ 0 \end{pmatrix} = L_A(\mathbf{i}) \quad \begin{pmatrix} -1 \\ -1 \end{pmatrix} = L_A(\mathbf{j}) \quad \begin{pmatrix} -4 \\ 3 \end{pmatrix} = L_A(\mathbf{k})$$

¹For us, example fields will almost always be $\mathbb{R}$ or $\mathbb{C}$ (sometimes $\mathbb{Q}$). If you've studied abstract algebra, you should also have encountered *extension fields* and *finite fields*. Prior experience with such examples is not necessary for these notes.

The example applies in general. Suppose $T \in \mathcal{L}(V, W)$ and that $\beta = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ and $\gamma = \{\mathbf{w}_1, \dots, \mathbf{w}_m\}$ are bases of V, W respectively. Then the *matrix of T with respect to β and γ* is the $m \times n$ matrix

$$[T]_{\beta}^{\gamma} = \left([T(\mathbf{v}_1)]_{\gamma} \cdots [T(\mathbf{v}_n)]_{\gamma} \right) \in M_{m \times n}(\mathbb{F}) \quad (\dagger)$$

whose j^{th} column is obtained by feeding the j^{th} basis vector of β to T before taking its co-ordinate vector with respect to γ . This fits naturally with the co-ordinate isomorphisms

$$T(\mathbf{v}) = \mathbf{w} \iff [T]_{\beta}^{\gamma}[\mathbf{v}]_{\beta} = [\mathbf{w}]_{\gamma}$$

There are two special cases when $V = W$:

- If $\beta = \gamma$, then we simply write $[T]_{\beta}$ instead of $[T]_{\beta}^{\beta}$.
- If $T = I$ is the identity map, then $Q_{\beta}^{\gamma} := [I]_{\beta}^{\gamma}$ is the *change of co-ordinate matrix* from β to γ .

Example. Let $T : P_2(\mathbb{R}) \rightarrow P_1(\mathbb{R})$ be the linear map defined by *differentiation*

$$T(a + bx + cx^2) = b + 2cx$$

The standard bases of $P_2(\mathbb{R})$ and $P_1(\mathbb{R})$ are, respectively, $\beta = \{1, x, x^2\}$ and $\gamma = \{1, x\}$. We compute the matrix of T with respect to these bases:

$$\begin{aligned} [T(1)]_{\gamma} &= [0]_{\gamma} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, & [T(x)]_{\gamma} &= [1]_{\gamma} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, & [T(x^2)]_{\gamma} &= [2x]_{\gamma} = \begin{pmatrix} 0 \\ 2 \end{pmatrix} \\ \implies [T]_{\beta}^{\gamma} &= \left([T(1)]_{\gamma} \ [T(x)]_{\gamma} \ [T(x^2)]_{\gamma} \right) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix} \end{aligned}$$

Written in co-ordinates and following $(\dagger)$, we observe the original linear map:

$$[T(a + bx + cx^2)]_{\gamma} \stackrel{(\dagger)}{=} [T]_{\beta}^{\gamma}[a + bx + cx^2]_{\beta} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} b \\ 2c \end{pmatrix} = [b + 2cx]_{\gamma}$$

Converting between linear maps and matrix multiplication is a central technique of linear algebra. As a quick pre-requisite test, work through the following:

Consider a second basis $\eta = \{1 + x, x + x^2, x^2 + 1\}$ of $P_2(\mathbb{R})$.

1. If $[p]_{\eta} = \begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix}$, what is the *polynomial* $p(x)$?
2. If $p(x) = 2 - 3x^2$, what is $[p]_{\eta}$?
3. Verify that η is indeed a basis of $P_2(\mathbb{R})$.
4. Show that $[T]_{\eta}^{\gamma} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & 2 \end{pmatrix}$.
5. Compute the change of co-ordinate matrix Q_{η}^{β} and check that the matrices of T are related by

$$[T]_{\eta}^{\gamma} = [T]_{\beta}^{\gamma} Q_{\eta}^{\beta}$$

1 Diagonalizability & the Cayley–Hamilton Theorem

1.1 Eigenvalues, Eigenvectors & Diagonalization (Review)

Diagonalization should be mostly familiar, though it is worth a more thorough review.

Definition 1.1. Suppose V is a vector space over $\mathbb{F}$ and $T \in \mathcal{L}(V)$. A non-zero $\mathbf{v} \in V$ is an *eigenvector* of T with *eigenvalue* $\lambda \in \mathbb{F}$ (together $(\lambda, \mathbf{v})$ are an *eigenpair*) if

$$T(\mathbf{v}) = \lambda \mathbf{v}$$

The *eigenspace* of an eigenvalue λ is the nullspace $E_\lambda := \mathcal{N}(T - \lambda I)$. Its *geometric multiplicity* is $\dim E_\lambda$. We say that T is *diagonalizable* if there exists a basis of eigenvectors: an *eigenbasis*.

Eigenvalues and Eigenvectors in finite dimensions

A simple diagonalizability criterion may be proved straightforwardly by induction.

Lemma 1.2. *Eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ with distinct eigenvalues are linearly independent.*

If $\dim_{\mathbb{F}} V = n$ and $T \in \mathcal{L}(V)$ has n distinct eigenvalues, then T is diagonalizable.

In finite dimensions, we often work with matrix examples or by choosing bases so as to view a linear maps as matrix multiplication. As such:

- The eigenvalues/vectors of a square matrix $A \in M_n(\mathbb{F})$ are precisely those of the linear map $L_A \in \mathcal{L}(\mathbb{F}^n)$, *left-multiplication* by A (that is, $L_A(\mathbf{x}) = A\mathbf{x}$).
- If $\dim_{\mathbb{F}} V = n$ and ϵ is a basis, then $T(\mathbf{v}) = \lambda \mathbf{v} \iff [T]_\epsilon[\mathbf{v}]_\epsilon = \lambda[\mathbf{v}]_\epsilon$. To be diagonalizable means there is a basis of eigenvectors $\beta = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ such that $[T]_\beta$ is *diagonal*

$$[T]_\beta = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & & \vdots \\ \vdots & & \ddots & 0 \\ 0 & \cdots & 0 & \lambda_n \end{pmatrix}$$

In such situations, there is systematic method for computing eigenvalues and eigenvectors:

1. (If necessary) Choose any basis ϵ of V and compute the matrix $A = [T]_\epsilon \in M_n(\mathbb{F})$.
2. Observe that

$$\begin{aligned} \lambda \in \mathbb{F} \text{ is an eigenvalue} &\iff \exists [\mathbf{v}]_\epsilon \in \mathbb{F}^n \setminus \{\mathbf{0}\} \text{ such that } A[\mathbf{v}]_\epsilon = \lambda[\mathbf{v}]_\epsilon \\ &\iff \exists [\mathbf{v}]_\epsilon \in \mathbb{F}^n \setminus \{\mathbf{0}\} \text{ such that } (A - \lambda I)[\mathbf{v}]_\epsilon = \mathbf{0} \\ &\iff \det(A - \lambda I) = 0 \end{aligned}$$

This last is a degree n polynomial equation whose roots are the eigenvalues.

3. For each eigenvalue λ_j , compute the eigenspace $E_{\lambda_j} = \mathcal{N}(T - \lambda_j I)$ to find the eigenvectors. Remember that E_{λ_j} is a subspace of the original vector space V , so translate back if necessary!

Definition 1.3. The *characteristic polynomial* of $T \in \mathcal{L}(V)$ is the degree- n polynomial²

$$p(t) := \det(T - tI)$$

The eigenvalues of T are precisely the solutions to the *characteristic equation* $p(t) = 0$.

Examples 1.4. 1. $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ has characteristic polynomial $p(t) = t^2 + 1 = (t + i)(t - i)$. Viewed as a linear map $L_A \in \mathcal{L}(\mathbb{R}^2)$, this matrix has *no eigenvalues* and *no eigenvectors*!

However, viewed as a linear map $L_A \in \mathcal{L}(\mathbb{C}^2)$, the matrix has two eigenvalues $\pm i$. Indeed

$$(A - iI)\mathbf{v} = \begin{pmatrix} -i & -1 \\ 1 & -i \end{pmatrix} \mathbf{v} \Rightarrow E_i = \text{Span} \begin{pmatrix} i \\ 1 \end{pmatrix}$$

Similarly $E_{-i} = \text{Span} \begin{pmatrix} -i \\ 1 \end{pmatrix}$. We therefore have an eigenbasis $\beta = \{ \begin{pmatrix} i \\ 1 \end{pmatrix}, \begin{pmatrix} -i \\ 1 \end{pmatrix} \}$ (of $\mathbb{C}^2$), with respect to which the linear map has a diagonal matrix:

$$[L_A]_\beta = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$$

2. Let $T \in \mathcal{L}(P_2(\mathbb{R}))$ be defined by

$$T(f)(x) = f(x) + (x - 1)f'(x)$$

With respect to the standard basis $\epsilon = \{1, x, x^2\}$, the linear map has a non-diagonal matrix:

$$A = [T]_\epsilon = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 2 & -2 \\ 0 & 0 & 3 \end{pmatrix} \Rightarrow p(t) = \det(A - tI) = (1 - t)(2 - t)(3 - t)$$

With three distinct eigenvalues, Lemma 1.2 says that T is diagonalizable. To find the eigenvectors, we compute the nullspaces.

$$\lambda_1 = 1: \mathbf{0} = (A - \lambda_1 I)[\mathbf{v}_1]_\epsilon = \begin{pmatrix} 0 & -1 & 0 \\ 0 & 1 & -2 \\ 0 & 0 & 2 \end{pmatrix} [\mathbf{v}_1]_\epsilon \Rightarrow [\mathbf{v}_1]_\epsilon \in \text{Span} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \Rightarrow E_1 = \text{Span}\{1\}$$

$$\lambda_2 = 2: A - \lambda_2 I = \begin{pmatrix} -1 & -1 & 0 \\ 0 & 0 & -2 \\ 0 & 0 & 1 \end{pmatrix} \Rightarrow [\mathbf{v}_2]_\epsilon \in \text{Span} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \Rightarrow E_2 = \text{Span}\{1 - x\}$$

$$\lambda_3 = 3: A - \lambda_3 I = \begin{pmatrix} -2 & -1 & 0 \\ 0 & -1 & -2 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow [\mathbf{v}_3]_\epsilon \in \text{Span} \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} \Rightarrow E_3 = \text{Span}\{1 - 2x + x^2\}$$

A sensible choice of eigenvectors produces an eigenbasis, with respect to which the linear map is necessarily diagonal

$$\beta = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \{1, 1 - x, 1 - 2x + x^2\} = \{1, 1 - x, (1 - x)^2\}$$

$$T(a + b(1 - x) + c(1 - x)^2) = a + 2b(1 - x) + 3c(1 - x)^2$$

$$[T]_\beta = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

²The determinant may be computed by considering the matrix of T with respect to *any* basis: $p(t)$ is independent of this choice (Exercise 8).

Conditions for diagonalizability of finite-dimensional operators

We borrow a little terminology from the theory of polynomials.

Definition 1.5. Let $\mathbb{F}$ be a field and $p(t)$ a polynomial with coefficients in $\mathbb{F}$.

1. Let $\lambda \in \mathbb{F}$ be a root: $p(\lambda) = 0$. The *algebraic multiplicity* $\text{mult}(\lambda)$ is the largest power of $\lambda - t$ to divide $p(t)$. Otherwise said, there exists some polynomial $q(t)$ such that

$$p(t) = (\lambda - t)^{\text{mult}(\lambda)} q(t) \quad \text{and} \quad q(\lambda) \neq 0$$

2. $p(t)$ *splits* over $\mathbb{F}$ if it factorizes completely into linear factors: $\exists a, \lambda_1, \dots, \lambda_k \in \mathbb{F}$ such that

$$p(t) = a(\lambda_1 - t)^{m_1} \dots (\lambda_k - t)^{m_k} \quad (*)$$

When $p(t)$ splits, the algebraic multiplicities sum to the degree n of the polynomial

$$n = m_1 + \dots + m_k$$

We are most interested when $p(t) = \det(T - tI)$ is the *characteristic polynomial* of a linear map $T \in \mathcal{L}(V)$. If such a polynomial splits, then (with notation as in $(*)$) two things are true:

- $a = 1$, by considering the degree n term of $p(t) = (-1)^n t^n + \dots$.
- $\lambda_1, \dots, \lambda_k$ are precisely the distinct eigenvalues of T .

Example (1.4.1, cont.). The field matters! For instance $p(t) = t^2 + 1 = (t - i)(t + i) = -(i - t)(-i - t)$ splits over $\mathbb{C}$ but not over $\mathbb{R}$. Its roots are plainly $\pm i$.

For the purposes of review, here is the main result. We will prove it in the next section.

Theorem 1.6. Let V be finite-dimensional. A linear map $T \in \mathcal{L}(V)$ is diagonalizable if and only if two conditions both hold:

1. Its characteristic polynomial splits over $\mathbb{F}$.
2. The geometric and algebraic multiplicities of each eigenvalue are equal; $\dim E_{\lambda_j} = \text{mult}(\lambda_j)$.

Example 1.7. The matrix $A = \begin{pmatrix} 3 & 1 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{pmatrix}$ is easily seen to have eigenvalues $\lambda_1 = 3$ and $\lambda_2 = 5$. Indeed

$$p(t) = (3 - t)^2(5 - t), \quad \text{mult}(3) = 2, \quad \text{mult}(5) = 1$$

$$E_3 = \text{Span} \left(\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \right), \quad E_5 = \text{Span} \left(\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right), \quad \dim E_3 = \dim E_5 = 1$$

This matrix is *non-diagonalizable* since $\dim E_3 = 1 \neq 2 = \text{mult}(3)$.

Everything prior to this *should* be review. If it feels very unfamiliar, revisit your notes from 121A, particularly sections 5.1 and 5.2 of the textbook.

Exercises 1.1. 1. For each matrix over $\mathbb{R}$: find its characteristic polynomial, its eigenvalues/spaces, and its algebraic and geometric multiplicities. Finally, decide if it is diagonalizable.

$$(a) A = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 3 & 1 & 0 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 3 \end{pmatrix} \quad (b) B = \begin{pmatrix} -1 & 6 & 0 & 0 \\ -2 & 6 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}$$

2. Suppose A is a *real* matrix with eigenpair $(\lambda, \mathbf{v})$. Show that the complex conjugates $(\bar{\lambda}, \bar{\mathbf{v}})$ also form an eigenpair.
3. Show that the characteristic polynomial of $A = \begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix}$ does not split over $\mathbb{R}$. Diagonalize A over the complex numbers $\mathbb{C}$.
4. Give an example of a 2×2 matrix whose entries are *rational numbers* and whose characteristic polynomial splits over $\mathbb{R}$, but not over $\mathbb{Q}$.
5. Diagonalize the matrix $A = \begin{pmatrix} 2i & 1 \\ 2 & 0 \end{pmatrix}$.
6. If a polynomial $p(t)$ splits over a field $\mathbb{F}$, explain why

$$\det T = \lambda_1^{\text{mult}(\lambda_1)} \dots \lambda_k^{\text{mult}(\lambda_k)}$$

where $\lambda_1, \dots, \lambda_k$ are the distinct eigenvalues of T .

7. Suppose $T \in \mathcal{L}(V)$ is invertible with eigenvalue λ . Prove that λ^{-1} is an eigenvalue of T^{-1} with the same eigenspace E_λ . If T is diagonalizable, prove that T^{-1} is also diagonalizable.

Your argument should work even if V is infinite-dimensional: no matrices!

8. If V is finite-dimensional and $T \in \mathcal{L}(V)$, we define

$$\det T := \det[T]_\beta$$

where β is any basis of V . Explain why the choice of basis does not matter: that is, if γ is any other basis of V , then $\det[T]_\gamma = \det[T]_\beta$.

9. Prove Lemma 1.2.
10. Suppose $A \in M_n(\mathbb{F})$ is diagonalizable with eigenbasis $\beta = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$. Prove that

$$A = MDM^{-1}$$

where $M = Q_\beta^\epsilon = [I]_\beta^\epsilon$ is the change of co-ordinate matrix (page 3) from β to the standard basis $\epsilon = \{\mathbf{e}_1, \dots, \mathbf{e}_n\}$, and $D = \text{diag}(\lambda_1, \dots, \lambda_n)$ is the diagonal matrix of eigenvalues.

1.2 Invariant Subspaces and the Cayley–Hamilton Theorem

The proof of Theorem 1.6 is facilitated by a new concept, of which eigenspaces are a special case.

Definition 1.8. Suppose $T \in \mathcal{L}(V)$. A subspace W of V is *T-invariant* if $T(W) \subseteq W$. In such a case, the restriction of T to W is the linear map

$$T_W : W \rightarrow W : \mathbf{w} \mapsto T(\mathbf{w})$$

As a shorthand, if $A \in M_n(\mathbb{F})$ then an L_A -invariant subspace $W \leq \mathbb{F}^n$ is simply called *A-invariant*.

Examples 1.9. 1. For any $T \in \mathcal{L}(V)$, the trivial subspace $\{0\}$ and the entirety of V are T -invariant.

2. Every eigenspace of T is T -invariant: if $\mathbf{v} \in E_\lambda$, then $T(\mathbf{v}) = \lambda \mathbf{v} \in E_\lambda$.

3. (Example 1.7, cont.) If $A = \begin{pmatrix} 3 & 1 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{pmatrix} \in M_3(\mathbb{R})$ then $W = \text{Span}\{\mathbf{i}, \mathbf{j}\} \leq \mathbb{R}^3$ is A -invariant:

$$A(x\mathbf{i} + y\mathbf{j}) = (3x + y)\mathbf{i} + 3y\mathbf{j} \in W$$

Alongside the trivial examples & eigenspaces, $\text{Span}\{\mathbf{i}, \mathbf{k}\} = E_3 \oplus E_5$ is also A -invariant.

To prove the diagonalization criterion (Theorem 1.6), we need to see how to factorize the characteristic polynomial. It turns out that factors of $p(t)$ correspond to T -invariant subspaces!

Example 1.10. $W = \text{Span}\{\mathbf{i}, \mathbf{j}\}$ is an invariant subspace of $A = \begin{pmatrix} 1 & 2 & 4 \\ 0 & 3 & 1 \\ 0 & 0 & 2 \end{pmatrix} \in M_3(\mathbb{R})$. With respect to the standard basis, the restriction of L_A to W has matrix $\begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$. The characteristic polynomial $p_W(t)$ of the restriction is a factor of the whole,

$$p(t) = (1 - t)(2 - t)(3 - t) = (2 - t)p_W(t)$$

Theorem 1.11. Suppose $T \in \mathcal{L}(V)$, that $\dim V$ is finite and that W is a T -invariant subspace of V . Then the characteristic polynomial $p_W(t)$ of the restriction T_W divides that of T .

The proof simply abstracts the approach of the example.

Proof. Extend a basis β_W of W to a basis β of V . Since $T(\mathbf{w}) \in \text{Span } \beta_W$ for each $\mathbf{w} \in W$, we see that the matrix of $[T]$ has block form

$$[T]_\beta = \left(\begin{array}{c|c} A & B \\ \hline O & C \end{array} \right) \Rightarrow p(t) = \det(A - tI) \det(C - tI) = p_W(t) \det(C - tI)$$

where $p_W(t)$ is the characteristic polynomial, and $A = [T_W]_{\beta_W}$ the matrix of the restriction T_W . ■

Corollary 1.12. If λ is an eigenvalue of T , then $T_{E_\lambda} = \lambda I_{E_\lambda}$ is a multiple of the identity. In particular:

1. The characteristic polynomial of the restriction T_{E_λ} is $p_\lambda(t) = (\lambda - t)^{\dim E_\lambda}$.
2. Since $p_\lambda(t)$ divides the characteristic polynomial of T , we see that $\dim E_\lambda \leq \text{mult}(\lambda)$.

We are now in a position to state and prove an extended version of Theorem 1.6.

Theorem 1.13. Suppose $\dim_{\mathbb{F}} V = n$ and that $T \in \mathcal{L}(V)$ has distinct eigenvalues $\lambda_1, \dots, \lambda_k$. The following are equivalent:

1. T is diagonalizable.
2. The characteristic polynomial splits over $\mathbb{F}$ and $\dim E_{\lambda_j} = \text{mult}(\lambda_j)$ for each j , indeed

$$p(t) = p_{\lambda_1}(t) \cdots p_{\lambda_k}(t) = (\lambda_1 - t)^{\dim E_{\lambda_1}} \cdots (\lambda_k - t)^{\dim E_{\lambda_k}}$$

$$3. \sum_{j=1}^k \dim E_{\lambda_j} = n$$

$$4. V = E_{\lambda_1} \oplus \cdots \oplus E_{\lambda_k}$$

Example 1.14. The data for the diagonalizable matrix $A = \begin{pmatrix} 7 & 0 & -12 \\ 0 & 1 & 0 \\ 2 & 0 & -3 \end{pmatrix}$ is shown in the table.

- $p(t) = (1 - t)^2(3 - t)$ splits
- $\mathbb{R}^4 = E_1 \oplus E_3$.
- $\beta = \left\{ \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ 0 \\ 1 \end{pmatrix} \right\}$ is an eigenbasis, with respect to which the map is diagonal $[L_A]_{\beta} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix}$

λ	1	3
$\text{mult}(\lambda)$	2	1
E_{λ}	$\text{Span} \left\{ \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}$	$\text{Span} \left(\begin{pmatrix} 3 \\ 0 \\ 1 \end{pmatrix} \right)$
$\dim E_{\lambda}$	2	1

Proof. (1 $\Rightarrow$ 2) If T is diagonalizable with eigenbasis β , then $[T]_{\beta}$ is diagonal. But then

$$p(t) = \det(\text{diag}(\lambda_1 - t, \dots, \lambda_k - t)) = (\lambda_1 - t)^{m_1} \cdots (\lambda_k - t)^{m_k}$$

splits and $\sum \text{mult}(\lambda_i) = n$. Plainly the cardinality n of an eigenbasis is *at most* $\sum \dim E_{\lambda_i}$. Corollary 1.12 supplies the second inequality in the following:

$$n \leq \sum \dim E_{\lambda_j} \leq \sum \text{mult}(\lambda_j) = n$$

All inequalities are necessarily *equalities*, whence $\dim E_{\lambda_j} = \text{mult}(\lambda_j)$ for each j .

(2 $\Rightarrow$ 3) $p(t)$ splits $\Rightarrow n = \sum \text{mult}(\lambda_j) = \sum \dim E_{\lambda_j}$

(3 $\Rightarrow$ 4) We sketch an induction argument.

Given $j \leq k$, assume $E_{\lambda_1} \oplus \cdots \oplus E_{\lambda_j}$ exists. If $(\lambda_{j+1}, \mathbf{v}_{j+1})$ is an eigenpair, then $\mathbf{v}_{j+1} \notin E_{\lambda_1} \oplus \cdots \oplus E_{\lambda_j}$ for otherwise Lemma 1.2 is contradicted. We conclude that $E_{\lambda_1} \oplus \cdots \oplus E_{\lambda_{j+1}}$ is a direct sum.

By induction, $E_{\lambda_1} \oplus \cdots \oplus E_{\lambda_k}$ exists. By assumption (condition 3) this direct sum has dimension $n = \dim V$ and therefore *equals* V .

(4 $\Rightarrow$ 1) For each j , choose a basis β_j of E_{λ_j} . Then $\beta := \beta_1 \cup \cdots \cup \beta_k$ is easily seen to be a basis of V (this is essentially condition 4, our assumption). Since every element of β is an eigenvector of T , we have an eigenbasis and so T is diagonalizable. ■

T-cyclic Subspaces and the Cayley–Hamilton Theorem

We introduce a general family of invariant subspaces before employing them to prove a startling result.

Definition 1.15. Let $T \in \mathcal{L}(V)$ and let $\mathbf{v} \in V$. The *T-cyclic subspace* generated by $\mathbf{v}$ is the span

$$\langle \mathbf{v} \rangle = \text{Span}\{\mathbf{v}, T(\mathbf{v}), T^2(\mathbf{v}), \dots\}$$

As you should expect by now, we say *A-cyclic* if $T = L_A$ is left-multiplication by a matrix.

Example (1.7, 1.9.3, cont.). Let $A = \begin{pmatrix} 3 & 1 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{pmatrix}$, and $\mathbf{v} = \mathbf{i} + \mathbf{k}$. It is easy to see that

$$A\mathbf{v} = 3\mathbf{i} + 5\mathbf{k}, \quad A^2\mathbf{v} = 9\mathbf{i} + 25\mathbf{k}, \quad \dots, \quad A^m\mathbf{v} = 3^m\mathbf{i} + 5^m\mathbf{k}$$

Plainly $\text{Span}\{\mathbf{i}, \mathbf{k}\} = \langle \mathbf{i} + \mathbf{k} \rangle$ is a *A-cyclic* subspace.

The proof of the following is left as an exercise.

Lemma 1.16. The *T-cyclic subspace* $\langle \mathbf{v} \rangle$ is the smallest *T-invariant* subspace of V containing $\mathbf{v}$. More specifically:

1. $\langle \mathbf{v} \rangle$ is *T-invariant*.
2. If $W \leq V$ is *T-invariant* and $\mathbf{v} \in W$, then $\langle \mathbf{v} \rangle \leq W$.
3. $\dim \langle \mathbf{v} \rangle = 1 \iff \mathbf{v}$ is an *eigenvector* of T .

We were lucky in the example that the general form $A^m\mathbf{v}$ was so clear. It is helpful to develop a general approach for identifying the dimension and a basis of a *T-cyclic* subspace.

Suppose $\mathbf{v} \neq \mathbf{0}$ and that $\langle \mathbf{v} \rangle$ has finite dimension.³ Let $k \geq 1$ be maximal such that the set

$$\{\mathbf{v}, T(\mathbf{v}), \dots, T^{k-1}(\mathbf{v})\}$$

is linearly independent.

- If k doesn't exist, the *infinite* linearly independent set $\{\mathbf{v}, T(\mathbf{v}), \dots\}$ contradicts $\dim \langle \mathbf{v} \rangle < \infty$.
- By the maximality of k , $T^k(\mathbf{v}) \in \text{Span}\{\mathbf{v}, T(\mathbf{v}), \dots, T^{k-1}(\mathbf{v})\}$. By induction this extends to

$$j \geq k \implies T^j(\mathbf{v}) \in \text{Span}\{\mathbf{v}, T(\mathbf{v}), \dots, T^{k-1}(\mathbf{v})\}$$

It follows that $\langle \mathbf{v} \rangle = \text{Span}\{\mathbf{v}, T(\mathbf{v}), \dots, T^{k-1}(\mathbf{v})\}$, and we've proved a useful criterion.

Theorem 1.17. Suppose $\mathbf{v} \neq \mathbf{0}$, then

$$\begin{aligned} \dim \langle \mathbf{v} \rangle = k &\iff \{\mathbf{v}, T(\mathbf{v}), \dots, T^{k-1}(\mathbf{v})\} \text{ is a basis of } \langle \mathbf{v} \rangle \\ &\iff k \text{ is maximal such that } \{\mathbf{v}, T(\mathbf{v}), \dots, T^{k-1}(\mathbf{v})\} \text{ is linearly independent} \\ &\iff k \text{ is minimal such that } T^k(\mathbf{v}) \in \text{Span}\{\mathbf{v}, T(\mathbf{v}), \dots, T^{k-1}(\mathbf{v})\} \end{aligned}$$

³Necessarily the situation if $\dim V < \infty$, whenever we are thinking about characteristic polynomials...

Examples 1.18. 1. According to the Theorem, in our ongoing Example could simply have noticed:

- $\mathbf{v} = \mathbf{i} + \mathbf{k}$ and $A\mathbf{v} = 3\mathbf{i} + 5\mathbf{k}$ are linearly independent.
- $A^2(\mathbf{i} + \mathbf{k}) = 9\mathbf{i} + 25\mathbf{k} \in \text{Span}\{\mathbf{v}, A\mathbf{v}\}$.

We'd then conclude that $\langle \mathbf{v} \rangle = \text{Span}\{\mathbf{v}, A\mathbf{v}\}$ has dimension 2.

2. Let $T(p(x)) = 3p(x) - p''(x)$ viewed as a linear map $T \in \mathcal{L}(P_2(\mathbb{R}))$ and consider the T-cyclic subspace generated by the polynomial $p(x) = x^2$

$$T(x^2) = 3x^2 - 2, \quad T^2(x^2) = T(3x^2 - 2) = 3(3x^2 - 2) - 6 = 9x^2 - 12, \quad \text{etc.}$$

Certainly $\{x^2, T(x^2)\}$ is linearly independent, but

$$T^2(x^2) = 9x^2 - 12 = -9x^2 + 6(3x^2 - 2) \in \text{Span}\{x^2, T(x^2)\}$$

We conclude that $\dim \langle x^2 \rangle = 2$. An alternative basis for $\langle x^2 \rangle$ is plainly $\{1, x^2\}$.

The crux of what follows is a simple observation: when restricting a linear map T to a T-cyclic subspace, the coefficients of the characteristic polynomial and the minimal linear combination in the Theorem coincide.

Continuing the Example, if $W = \langle x^2 \rangle$ and $\beta_W = \{x^2, T(x^2)\} = \{x^2, 3x^2 - 2\}$, then

$$[T_W]_{\beta_W} = \begin{pmatrix} 0 & -9 \\ 1 & 6 \end{pmatrix} \Rightarrow p_W(t) = t^2 - 6t + 9$$

Lemma 1.19. Let $T \in \mathcal{L}(V)$ and suppose $W = \langle \mathbf{w} \rangle$ has $\dim W = k$ and basis

$$\beta_W = \{\mathbf{w}, T(\mathbf{w}), \dots, T^{k-1}(\mathbf{w})\}$$

in accordance with Theorem 1.17. Then:

1. If $T^k(\mathbf{w}) + a_{k-1}T^{k-1}(\mathbf{w}) + \dots + a_0\mathbf{w} = \mathbf{0}$, then the characteristic polynomial of T_W is

$$p_W(t) = (-1)^k (t^k + a_{k-1}t^{k-1} + \dots + a_1t + a_0)$$

2. $p_W(T_W) = 0$ is the zero map on W .

Proof. 1. This is a tedious exercise: just expand the relevant determinant row by row...

2. Write $S \in \mathcal{L}(V)$ for the linear map

$$S := p_W(T) = (-1)^k (T^k + a_{k-1}T^{k-1} + \dots + a_0I)$$

Part 1 says $S(\mathbf{w}) = \mathbf{0}$. Since S is a polynomial in T , it commutes with all powers of T :

$$\forall j, S(T^j(\mathbf{w})) = T^j(S(\mathbf{w})) = \mathbf{0}$$

Since S is zero on the basis β_W of W , we see that S_W is the zero function. ■

With a little sneakiness, we can drop the W 's entirely and observe an intimate relation between a linear map and its characteristic polynomial.

Theorem 1.20 (Cayley–Hamilton). *If V is finite-dimensional, then $T \in \mathcal{L}(V)$ satisfies its characteristic polynomial: $p(T) = 0$.*

Proof. Let $\mathbf{w} \in V$ and consider the cyclic subspace $W = \langle \mathbf{w} \rangle$ generated by $\mathbf{w}$. By Theorem 1.11,

$$p(t) = q_W(t)p_W(t)$$

for some polynomial q_W . Lemma 1.19 says that $(p_W(T))(\mathbf{w}) = \mathbf{0}$, whence

$$(p(T))(\mathbf{w}) = \mathbf{0}$$

Since we can apply this reasoning to *any* $\mathbf{w} \in V$, we conclude that $p(T)$ is the zero function. ■

Examples 1.21. 1. $A = \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix}$ has $p(t) = t^2 - 6t + 5$ and we confirm:

$$A^2 - 6A = \begin{pmatrix} 7 & 6 \\ 18 & 19 \end{pmatrix} - 6 \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix} = -5I$$

It is a strange approach for this matrix, but the characteristic equation can be used to calculate the inverse of A :

$$A^2 - 6A + 5I = 0 \implies A(A - 6I) = -5I \implies A^{-1} = \frac{1}{5}(6I - A) = \frac{1}{5} \begin{pmatrix} 4 & -1 \\ -3 & 2 \end{pmatrix}$$

2. We use Cayley–Hamilton to compute A^4 when

$$A = \begin{pmatrix} 2 & -1 & \frac{8}{3} \\ 0 & 1 & -6 \\ 0 & 0 & 2 \end{pmatrix}$$

The characteristic polynomial is

$$p(t) = (2 - t)^2(1 - t) = 4 - 8t + 5t^2 - t^3$$

whence

$$\begin{aligned} A^4 &= AA^3 = A(5A^2 - 8A + 4I) \\ &= 5A^3 - 8A^2 + 4A = 5(5A^2 - 8A + 4I) - 8A^2 + 4A \\ &= 17A^2 - 36A + 20I = 17 \begin{pmatrix} 4 & -3 & \frac{50}{3} \\ 0 & 1 & -18 \\ 0 & 0 & 4 \end{pmatrix} - 36 \begin{pmatrix} 2 & -1 & \frac{8}{3} \\ 0 & 1 & -6 \\ 0 & 0 & 2 \end{pmatrix} + 20 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 16 & -15 & \frac{562}{3} \\ 0 & 1 & -90 \\ 0 & 0 & 16 \end{pmatrix} \end{aligned}$$

3. (Example 1.4.2, cont.) The linear map $T(f(x)) = f(x) + (x-1)f'(x)$ has

$$p(t) = (1-t)(2-t)(3-t) = -t^3 + 6t^2 - 11t + 6$$

By Cayley–Hamilton, $T^3 = 6T^2 - 11T + 6I$. You can check this explicitly, after first computing

$$T^2(f(x)) = f(x) + 3(x-1)f'(x) + (x-1)^2f''(x), \text{ etc.}$$

We can also apply Cayley–Hamilton to simplify higher powers of T and even to compute the inverse:

$$\begin{aligned} I &= \frac{1}{6}(T^3 - 6T^2 + 11T) \implies T^{-1} = \frac{1}{6}(T^2 - 6T + 11I) \\ &\implies T^{-1}(f(x)) = f(x) - \frac{1}{2}(x-1)f'(x) + \frac{1}{6}(x-1)^2f''(x) \end{aligned}$$

This should feel very strange: the inverse of T involves more derivatives!

Exercises 1.2. 1. Given $A = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 4 \\ 0 & 0 & 2 \end{pmatrix}$ find the A -cyclic subspace of $\mathbb{R}^3$ generated by $\mathbf{k} = \mathbf{e}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$.

2. Given $A = \begin{pmatrix} 1 & 2 & 4 \\ 0 & 3 & 1 \\ 0 & 0 & 2 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$, compute $A(\mathbf{v})$ and $A^2(\mathbf{v})$. Hence describe the A -cyclic subspace $\langle \mathbf{v} \rangle$ and its dimension.

3. Given $A = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 3 & 1 & 0 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 3 \end{pmatrix}$, find *two distinct* A -invariant subspaces $W \leq \mathbb{R}^4$ such that $\dim W = 3$.

4. Suppose W and X are T -invariant subspaces of V . Prove that the sum

$$W + X = \{\mathbf{w} + \mathbf{x} : \mathbf{w} \in W, \mathbf{x} \in X\}$$

is also T -invariant.

5. Prove all three parts of Lemma 1.16.

6. Give an example of an infinite-dimensional vector space V , a linear map $T \in \mathcal{L}(V)$, and a vector $\mathbf{v}$ such that $\langle \mathbf{v} \rangle = V$. Explain your answer.

7. Consider $\beta = \{\sin x, \cos x, 2x \sin x, 3x \cos x\}$ and $T = \frac{d}{dx} \in \mathcal{L}(\text{Span } \beta)$. Plainly the subspace $W := \text{Span}\{\sin x, \cos x\}$ is T -invariant. Compute the matrices $[T]_\beta$ and $[T_W]_{\beta_W}$ and observe that

$$p(t) = (p_W(t))^2$$

8. Verify explicitly that $A = \begin{pmatrix} 2 & 3 \\ 0 & 2 \end{pmatrix}$ satisfies its characteristic polynomial, in accordance with Cayley–Hamilton.

9. Check the details of Example 1.21.3. Moreover:

(a) Evaluate T^4 as a linear combination of I , T and T^2 .

(b) Verify the evaluation of $T^{-1}(f(x))$.

10. Suppose a, b are constants with $a \neq 0$ and define $T(f(x)) = af(x) + bf'(x)$.

(a) Find an expression for the inverse $T^{-1}(f(x))$ if $T \in \mathcal{L}(P_1(\mathbb{R}))$

(b) Find an expression for the inverse $T^{-1}(f(x))$ if $T \in \mathcal{L}(P_2(\mathbb{R}))$

Your answers should be written in terms of f and its derivatives.

11. Let $T(f)(x) = f'(x) + \frac{1}{x} \int_0^x f(t) dt$ be a linear map $T \in \mathcal{L}(P_2(\mathbb{R}))$.

(a) Find the characteristic polynomial of T and identify its eigenspaces. Is T diagonalizable?

(b) Find $a, b, c \in \mathbb{R}$ such that $T^3 = aT^2 + bT + cI$.

(c) What are $\dim \mathcal{L}(P_2(\mathbb{R}))$ and $\dim \text{Span}\{T^k : k \in \mathbb{N}_0\}$? Explain.

12. If $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ has non-zero determinant, use Cayley–Hamilton to obtain the usual expression for A^{-1} .

13. Recall Examples 1.7, 1.9.3 and 1.18.1 with $A = \begin{pmatrix} 3 & 1 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{pmatrix}$.

(a) If $\mathbf{v} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$, show that $\det(\mathbf{v}, A\mathbf{v}, A^2\mathbf{v}) = -4y^2z$

(b) Hence determine *all* A -cyclic subspaces of $\mathbb{R}^3$.

14. (a) Consider Example 1.18.2 where $T \in \mathcal{L}(P_2(\mathbb{R}))$ is defined by $T(p(x)) = 3p(x) - p''(x)$. Prove that all T -cyclic subspaces have dimension ≤ 2 .

(b) What if we instead consider $S \in \mathcal{L}(P_2(\mathbb{R}))$ defined by $S(p(x)) = 3p(x) - p'(x)$?

15. We prove part 1 of Lemma 1.19.

(a) Explain why the matrix of T_W with respect to the basis β_W is

$$[T_W]_{\beta_W} = \begin{pmatrix} 0 & 0 & 0 & \cdots & 0 & -a_0 \\ 1 & 0 & 0 & & 0 & -a_1 \\ 0 & 1 & 0 & & 0 & -a_2 \\ \vdots & & & \ddots & & \vdots \\ 0 & 0 & 0 & & 0 & -a_{k-2} \\ 0 & 0 & 0 & \cdots & 1 & -a_{k-1} \end{pmatrix} \in M_k(\mathbb{F})$$

(b) Compute the characteristic polynomial $p_W(t) = \det([T_W]_{\beta_W} - tI_k)$ by expanding the determinant along the first row.